

FWA vs GR: Numerical Validation Plan (5 Tests, 1 Parameter)

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1. Core Model

1.1. Potential:

$$\Phi(r) = -GM/r + \lambda \ln(r/r_0)$$

1.2. Derivative:

$$d\Phi/dr = GM/r^2 + \lambda/r$$

1.3. Asymptotic Velocity:

$$v^2(r) = r \cdot d\Phi/dr = GM/r + \lambda$$

As $r \rightarrow \infty$: $v(r) \rightarrow \sqrt{\lambda}$

1.4. Parameter Identification: $\lambda = v_{\text{flat}}^2$

1.5. Dimension Check: • Φ : m^2/s^2 • λ : m^2/s^2 • r_0 : m • v^2 : m^2/s^2

T001 — Galaxy Rotation Curve

Observable: $v(r)$

FWA Prediction: $v^2(r) = GM/r + \lambda$

Asymptotic Behavior: $v_{\text{flat}} = \sqrt{\lambda}$

Milky Way Example: $v_{\text{flat}} = 220 \text{ km/s} = 2.2 \times 10^5 \text{ m/s} \Rightarrow \lambda = 4.84 \times 10^{10} \text{ m}^2/\text{s}^2$

Status: Closed-form. Ready for numerical integration.

T002 — Perihelion Precession

GR Contribution: $\Delta\phi_{\text{GR}} = 6\pi GM / [c^2 a(1-e^2)]$

FWA Correction (Dimensional Placeholder): $\delta\phi_{\text{FWA}} \sim C \cdot (\lambda/GM) \cdot a(1-e^2)$

Notes: Coefficient C must be derived from full perturbation theory. Expression is dimensionally correct but not yet derived. Must be negligible for Mercury if FWA uses λ from MW.

Falsification Criterion: If $\delta\phi_{\text{FWA}}$ exceeds observational error $\rightarrow$ FWA falsified.

Status: Heuristic. Needs full derivation.

T003 — Gravitational Lensing

General Expression: $\alpha(b) = (1/c^2) \int [\nabla_{\perp} \Phi] dz$ from $-\infty$ to ∞

Derived FWA Form: $\alpha(b) = 4GM/(c^2 b) + 2\pi\lambda/(c^2 b)$

Assumptions: Spherical symmetry, Thin lens approximation.

Milky Way Example: $b = 8 \text{ kpc} = 2.47 \times 10^{20} \text{ m}$, $\lambda = 4.84 \times 10^{10} \text{ m}^2/\text{s}^2$

Status: Closed-form. Ready for numerical evaluation.

T004 — ISCO Radius

Newtonian Check: $d/dr [r^2(d\Phi/dr)] = \lambda \neq 0$

Conclusion: No ISCO in Newtonian FWA. Requires relativistic FWA metric.

Status: Future Work. Do not use for claims.

T005 — Jet Power

Ansatz: $P_{\text{jet}} \sim k \cdot |\nabla\Phi|_{\text{core}}^2 \cdot f(L_u, \text{node})$

Gradient: $|\nabla\Phi| \sim GM/r_{\text{core}}^2 + \lambda/r_{\text{core}}$

Notes: Phenomenological ansatz. Must define $f(L_u, \text{node})$ from FWA dynamics before fitting.

Status: Order-of-magnitude only.

3. Numerical Example: Milky Way

Inputs: $v_{\text{flat}} = 220 \text{ km/s}$, $b = 8 \text{ kpc}$, $M(b) \approx 10^{11} M_{\odot}$ (placeholder)

Outputs:

$$\lambda = 4.84 \times 10^{10} \text{ m}^2/\text{s}^2$$

$$v(r) = \sqrt{GM/r + \lambda}$$

$$\alpha_{\text{FWA}}(b) = \alpha_{\text{GR}}(b) + 2\pi\lambda/(c^2 b)$$

Success Criterion: FWA lensing term must reproduce DM-like contribution without violating T002.

4. Code Plan

Language: Python

Modules:

- orbit_integrator.py — compute $v(r)$
- lensing_calculator.py — compute $\alpha(b)$
- lambda_scanner.py — scan λ vs observables

Repository: github.com/Igorggghhh/FWA-Ontology--v3

5-7. Scope, Statements & Actions

5. Scientific Statement: FWA proposes a single dynamical alternative to geometric GR. It is validated or falsified by five independent tests using one parameter λ .

6. Claim Scope: Compare FWA vs GR on observables. Do not state "GR is wrong". State only: "FWA passes/fails test T00X".

7. Next Action: 1. Implement T001 and T003 in Python. | 2. Run Milky Way test. | 3. Publish results in GitHub + Zenodo v3.